

B.Tech. Degree IV Semester Regular/Supplementary Examination June 2022

ME 19-205-0403 MECHATRONICS
(2019 Scheme)

Time: 3 Hours

Maximum Marks: 60

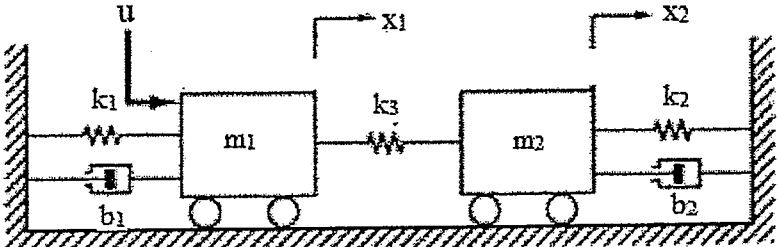
Course Outcome

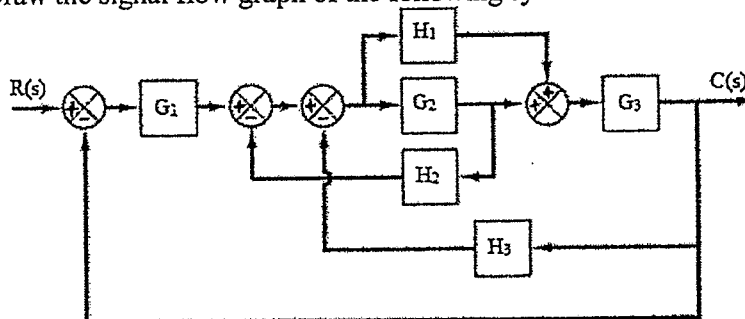
On successful completion of the course, the students will be able to:

- CO1: **Understand** the basics of sensors and transducers and different mechatronics systems.
 CO2: **Model** first and second order systems using block diagrams and governing equations.
 CO3: **Analyse** first and second order systems for their time and frequency domain responses and stability.
 CO4: **Design** mechatronics system for simple application.
 CO5: **Use** software tools for modelling and simulation of mechatronic systems.
 Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 – Analyze, L5 – Evaluate, L6 – Create
 PO – Programme Outcome

PART A

(Answer **ALL** questions)

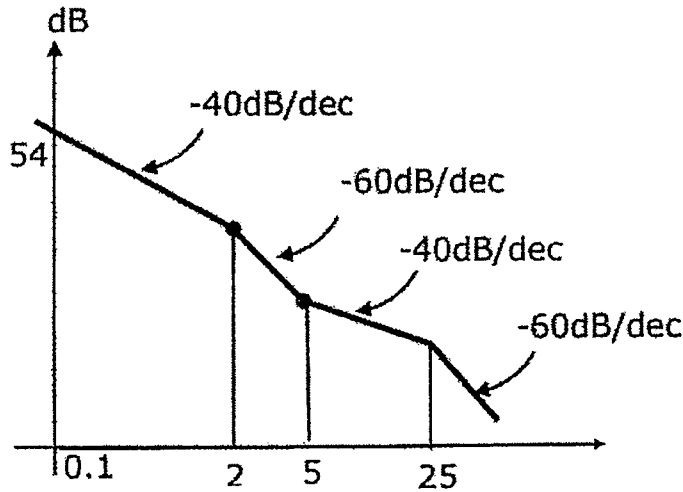
- | I. | (8 × 3 = 24) | Marks | BL | CO | PO |
|-----|---|-------|----|----|----------------|
| (a) | What is a Ratchet and pawl mechanism? Explain its working principle. | 3 | L1 | 1 | 1,2,3,4,8,12 |
| (b) | Draw the symbol of a 5 port 2 position direction control valve which is actuated by solenoids on both sides and explain its working. | 3 | L2 | 1 | 1,2,3,4,8,12 |
| (c) |  | 3 | L3 | 2 | 1,2,3,4,5,8,12 |
| (d) | For the mechanical system shown above, determine the force-voltage analogous system.
Draw the signal flow graph of the following system. | 3 | L3 | 2 | 1,2,3,4,5,8,12 |



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- (e) The asymptotic approximation of $20 \log(|G(j\omega)|)$ versus ω of a minimum phase system with real poles and one zero is shown below. Determine the transfer function of the system.



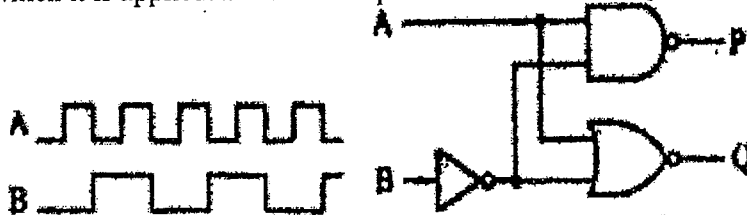
- (f) The open loop transfer function of a unity negative feed system is $G(s) = \frac{K}{s^2 + 4s - 5}$. Find the range of K for which the system will be stable.
- (g) Differentiate between a Microprocessor and a Microcontroller.
- (h) Represent the following Boolean algebra equations in a PLC logic ladder diagram.

$$Z = A \cdot \overline{B} \cdot C + \overline{D} + E$$

PART B

(4 × 12 = 48)

- II. (a) The figure shows the input signals A and B and the gate system to which it is applied. Draw the output waveforms P and Q.



- (b) Explain the working principle of an AC stepper motor and obtain the transfer function.

OR

- III. (a) List out the displacement sensors and explain the working principle of LVDT.
- (b) A 24 V lamp is to be switched ON when a push button switch is pressed. It has to remain ON even after the push button is released. Design a circuit for this using a 24 V, 2-pole relay.

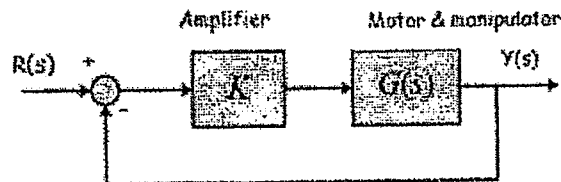
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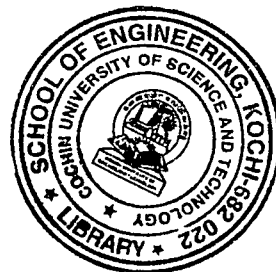
- IV. The use of laser for surgery requires high accuracy for the position and velocity responses (or step and ramp inputs). The figure below shows the block diagram of a laser surgery system which uses a dc motor and a manipulator for the laser. The transfer function for the

$$\text{motor and manipulator is } G(s) = \frac{K}{s(s+10)(s+5)}$$

(This data is common for Questions IV, V, VI, and VII)



- | | | | | | |
|-----------|---|---|----|---|----------------|
| (i) | What is the range of values for K for which the system will be stable? | 6 | L4 | 3 | 1,2,3,4,5,8,12 |
| (ii) | Obtain the state space representation of the system for K=50. | 6 | | | |
| OR | | | | | |
| V. | (i) Referring to the system given in Question IV, what should be the amplifier gain K so that the steady state error for a ramp input $r(t) = At$ (where $A = 1 \text{ mm/s}$) should be less than or equal to 0.1 mm while remaining in stable condition? | 4 | L4 | 3 | 1,2,3,4,5,8,12 |
| | (ii) For that value of K found above, find the closed loop poles of the system. | 2 | | | |
| | (iii) Can the system be approximated as a second order system? Justify your answer. | 2 | | | |
| | (iv) If yes, determine the percentage overshoot and the 2% settling time of the system. | 4 | | | |
| VI. | (i) Plot the root locus of the system given in Question IV. | 6 | L4 | 3 | 1,2,3,4,5,8,12 |
| | (ii) For an amplifier gain K= 100, find the closed loop poles. | 2 | | | |
| | (iii) Determine the transfer function of the PD controller for the above if the percentage overshoot is to be limited to 25% without any change in the settling time. | 4 | L5 | 4 | |
| OR | | | | | |
| VII. | (i) Draw the frequency response curves (Bode diagram) for the system given in Question IV for an amplifier gain K=100 | 6 | L4 | 3 | 1,2,3,4,5,8,12 |
| | (ii) Identify the gain margin and phase margin. Can you comment on the stability of the system from the margins? | 2 | L5 | 4 | |
| | (iii) How much change can be made to the amplifier gain before the system becomes unstable? | 2 | | | |
| | (iv) Determine the value of K_v from the Bode diagram and evaluate the velocity error constant. | 2 | | | |



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VIII.	(a)	Explain the Fundamentals of real-time processing in automation and control system.	6	L2	2	1,2,3,4,5,8,12
	(b)	Draw the block diagram and explain how an automatic car parking system is implemented. List out various types of sensors in this system	6	L5	4	
OR						
IX.	(a)	Draw the architecture of Programmable logic controller and explain.	6	L3	1	1,2,3,4,8,12
	(b)	A water pump for filling a tank is started using a spring-return push-button switch. When a sensor detects that the tank is full, the pump is stopped. Then the water is heated for 300 seconds using a heater. Draw a ladder diagram to represent this.	6	L5	4	1,2,3,4,5,8,12

Blooms's Taxonomy Levels

L1 – 7.5%, L2 - 10%, L3 - 20%, L4 - 35%, L5 – 27.5%
